

Supplementary Material to the manuscript entitled 'Pace of life and mobility as key factors to survive in farmland – relationships between functional traits of diurnal Lepidoptera and landscape structure'.

Table S1. Traits of butterflies and burnet moths used in the analysis with species names abbreviations, the number of sites with occurrences (Occ.) and total abundance (Ab.) of all 104 species.

Species	Mobility	Larval diet	Overwintering stage	Voltinism	HA	HAS	Abbreviation	Occ.	Ab.
<i>Adscita geryon</i>	2	M	Larva	1	1	2	Ads.ger	1	1
<i>Adscita statices</i>	2	O	Larva	1	2	3	Ads.sta	6	13
<i>Aglais io</i>	3	M	Adult	1.5	3	3	Agl.io	66	344
<i>Aglais urticae</i>	3	M	Adult	2	2	2	Agl.urt	91	586
<i>Anthocharis cardamines</i>	2	O	Pupa	1	3	5	Ant.car	18	37
<i>Apatura ilia</i>	2.5	O	Larva	1.5	1	2	Apa.ili	4	4
<i>Apatura iris</i>	2.5	O	Larva	1	1	1	Apa.iri	4	6
<i>Aphantopus hyperantus</i>	2	O	Larva	1	4	8	Aph.hyp	70	1368
<i>Aporia crataegi</i>	2.5	O	Larva	1	3	5	Apo.cra	8	59
<i>Araschnia levana</i>	2.5	M	Pupa	2.5	2	3	Ara.lev	33	132
<i>Argynnis adippe</i>	2.5	NO	Egg	1	2	4	Arg.adi	13	61
<i>Argynnis aglaja</i>	2.5	P	Larva	1	1	2	Arg.agl	12	65
<i>Argynnis niobe</i>	2.5	NO	Egg	1	1	2	Arg.nio	1	5
<i>Argynnis paphia</i>	2.5	P	Larva	1	1	1	Arg.pap	43	406
<i>Aricia agestis</i>	1.5	P	Larva	2.5	2	3	Ari.age	18	34
<i>Aricia artaxerxes</i>	1.5	P	Larva	1	1	2	Ari.art	2	6
<i>Aricia eumedon</i>	1	NO	Larva	1	3	6	Ari.eum	4	37
<i>Boloria dia</i>	2	NO	Larva	2.5	2	2	Bol.dia	17	198
<i>Boloria eunomia</i>	1	M	Larva	1	1	2	Bol.eun	4	49
<i>Boloria euphrosyne</i>	2	NO	Larva	1	2	3	Bol.eup	8	28
<i>Boloria selene</i>	1.5	NO	Larva	1.5	3	5	Bol.sel	9	19
<i>Boloria titania</i>	2	P	Larva	1	2	3	Bol.tit	2	12
<i>Brenthis ino</i>	1	O	Egg	1	1	3	Bre.ino	16	141
<i>Callophrys rubi</i>	2	P	Pupa	1	3	5	Cal.rub	4	9
<i>Carcharodus alceae</i>	3	O	Larva	2	2	4	Car.alc	3	7
<i>Carterocephalus palaemon</i>	1	O	Larva	1	3	6	Car.pal	10	19
<i>Celastrina argiolus</i>	2.5	P	Pupa	2.5	1	2	Cel.arg	8	10
<i>Coenonympha arcania</i>	2	O	Larva	1.5	2	4	Coe.arc	13	169
<i>Coenonympha glycerion</i>	1.5	O	Larva	1	2	2	Coe.gly	5	34
<i>Coenonympha pamphilus</i>	2	O	Larva	3	3	5	Coe.pam	87	1548
<i>Coenonympha tullia</i>	1	P	Larva	1	1	1	Coe.tul	2	10
<i>Colias alfacariensis</i>	3	O	Larva	3	2	3	Col.alf	7	15
<i>Colias croceus</i>	4	O	Larva	2	2	5	Col.cro	3	4
<i>Colias hyale</i>	3	O	Larva	2.5	2	5	Col.hya	30	88
<i>Cupido argiades</i>	2.5	O	Larva	3	4	6	Cup.arg	36	126

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<i>Cupido minimus</i>	1	O	Larva	2	2	3	Cup.min	4	25
<i>Erebia aethiops</i>	2	O	Larva	1	1	2	Ere.aet	1	13
<i>Erebia ligea</i>	2	O	Egg	1	1	1	Ere.lig	3	13
<i>Erebia medusa</i>	2	O	Larva	1	3	6	Ere.med	14	69
<i>Erebia meolans</i>	2	NO	Larva	1	2	2	Ere.meo	3	56
<i>Erynnis tages</i>	2	O	Larva	1.5	2	4	Ery.tag	15	42
<i>Glauropsyche alexis</i>	1	O	Pupa	1	2	3	Gla.ale	1	1
<i>Gonepteryx rhamni</i>	2	O	Adult	1	1	2	Gon.rha	50	235
<i>Hamearis lucina</i>	1	NO	Pupa	1	3	4	Ham.luc	2	2
<i>Issoria lathonia</i>	3	NO	Pupa	3	3	5	Iss.lat	23	71
<i>Jordanita globulariae</i>	2	O	Larva	1	3	5	Jor.glo	1	1
<i>Lasiommata maera</i>	2	O	Larva	1.5	2	4	Las.mae	4	19
<i>Lasiommata megera</i>	2	O	Larva	2.5	2	4	Las.meg	15	44
<i>Leptidea sp.</i>	1	O	Pupa	2.5	3	5	Lep.sp	48	298
<i>Limenitis camilla</i>	2	NO	Larva	1.5	1	1	Lim.cam	16	41
<i>Limenitis populi</i>	2	NO	Larva	1	1	1	Lim.pop	1	1
<i>Lycaena dispar</i>	2	NO	Larva	2	3	4	Lyc.dis	1	4
<i>Lycaena hippothoe</i>	2	NO	Larva	1	2	3	Lyc.hip	2	2
<i>Lycaena phlaeas</i>	3	NO	Larva	2.5	2	5	Lyc.phl	32	55
<i>Lycaena tityrus</i>	2	NO	Larva	2	3	6	Lyc.tit	24	67
<i>Maculineaalcon XER</i>	1	M	Larva	1	1	2	Mac.alc	1	7
<i>Maculinea nausithous</i>	1	M	Larva	1	1	1	Mac.nau	4	8
<i>Maculinea teleius</i>	1	M	Larva	1	1	1	Mac.tel	1	2
<i>Maniola jurtina</i>	2	O	Larva	1	4	6	Man.jur	87	5739
<i>Melanargia galathea</i>	2	O	Larva	1	2	4	Mel.gal	70	3248
<i>Melitaea athalia</i>	2	P	Larva	1	3	4	Mel.ath	12	79
<i>Melitaea aurelia</i>	1	NO	Larva	1	1	2	Mel.aur	2	7
<i>Melitaea britomartis</i>	1	P	Larva	1	2	2	Mel.bri	5	95
<i>Melitaea cinxia</i>	2	P	Larva	1.5	1	2	Mel.cin	5	27
<i>Melitaea diamina</i>	2	P	Larva	1	3	4	Mel.dia	10	34
<i>Melitaea didyma</i>	2	P	Larva	1.5	2	3	Mel.did	3	70
<i>Nymphalis antiopa</i>	2.5	P	Adult	1	1	2	Nym.ant	3	3
<i>Nymphalis polychloros</i>	2.5	P	Adult	1	1	2	Nym.pol	4	4
<i>Ochlodes sylvanus</i>	2	O	Larva	1	4	7	Och.syl	57	212
<i>Papilio machaon</i>	3	P	Pupa	2	3	7	Pap.mac	26	50
<i>Pararge aegeria</i>	2	O	Pupa	2.5	1	2	Par.aeg	21	63
<i>Pieris brassicae</i>	3	O	Pupa	3	2	5	Pie.bra	54	346
<i>Pieris napi</i>	3	O	Pupa	3.5	4	8	Pie.nap	78	688
<i>Pieris rapae</i>	3	O	Pupa	4	2	5	Pie.rap	94	2457
<i>Plebeius argus</i>	1.5	O	Egg	1.5	2	3	Ple.arg	5	49
<i>Plebeius argyrognomon</i>	1.5	O	Egg	2.5	2	3	Pl.argy	5	68
<i>Polygonia c-album</i>	3	P	Adult	2	1	2	Pol.c-a	26	47
<i>Polyommatus bellargus</i>	2	O	Larva	2.5	2	3	Pol.bel	12	142
<i>Polyommatus coridon</i>	1.5	O	Egg	1	2	4	Pol.cor	8	179

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<i>Polyommatus daphnis</i>	1.5	O	Egg	1	1	1	Pol.dap	1	1
<i>Polyommatus icarus</i>	2	O	Larva	2.5	3	7	Pol.ica	91	1181
<i>Polyommatus semiargus</i>	1.5	O	Larva	2.5	3	5	Pol.sem	58	513
<i>Polyommatus thersites</i>	2	NO	Larva	2	1	2	Pol.the	4	122
<i>Pyrgus armoricanus</i>	2	NO	Larva	2.5	3	4	Pyr.arm	1	1
<i>Pyrgus malvae</i>	2	O	Pupa	1.5	4	6	Pyr.mal	9	19
<i>Pyronia tithonus</i>	1.5	O	Larva	1	3	5	Pyr.tit	5	95
<i>Satyrium acaciae</i>	1	M	Egg	1	2	3	Sat.aca	1	1
<i>Satyrium pruni</i>	2.5	NO	Egg	1	1	2	Sat.pru	11	35
<i>Satyrium spini</i>	2	NO	Egg	1	2	4	Sat.spi	2	3
<i>Satyrium w-album</i>	1.5	NO	Egg	1	1	2	Sat.w-a	3	4
<i>Spialia sertorius</i>	2	M	Larva	2	2	3	Spi.ser	4	13
<i>Thecla betulae</i>	2	NO	Egg	1	1	2	The.bet	1	1
<i>Thymelicus lineola</i>	2	O	Egg	1	4	7	Thy.lin	70	1073
<i>Thymelicus sylvestris</i>	2	O	Larva	1	3	5	Thy.syl	39	245
<i>Vanessa atalanta</i>	4	M	Adult	3.5	4	4	Van.ata	45	72
<i>Zygaena carniolica</i>	2	O	Larva	1	1	2	Zyg.car	1	15
<i>Zygaena ephialtes</i>	2	O	Larva	1	3	5	Zyg.eph	2	8
<i>Zygaena filipendulae</i>	2	O	Larva	1	4	7	Zyg.fil	17	86
<i>Zygaena lonicerae</i>	2	O	Larva	1	2	4	Zyg.lon	1	3
<i>Zygaena loti</i>	2	O	Larva	1	2	3	Zyg.lot	11	113
<i>Zygaena purpuralis</i>	2	NO	Larva	1	2	4	Zyg.pur	4	99
<i>Zygaena transalpina</i>	2	O	Larva	1	2	4	Zyg.tra	4	28
<i>Zygaena trifolii</i>	2	O	Larva	1	1	3	Zyg.tri	2	25
<i>Zygaena viciae</i>	2	O	Larva	1	4	6	Zyg.vic	4	21

Table S2. Correlation tests between mobility scores used in our study and in previous ones.

Reference dataset	Geographic region	No. overlapping species	Correlation coefficient	<i>p</i> -value
Bink (1992)	NW Europe	89	Pearson's $r = 0.75$	$<< 0.001$
Cowley <i>et al.</i> (2001)	UK	43	Spearman's $\rho = 0.82$	$<< 0.001$
Komonen <i>et al.</i> (2004)	Finland	53	Pearson's $r = 0.52$	$<< 0.001$

Figure S1. Map of an exemplary arable site at Oberkessach, Germany. 340 m a.s.l. A: the red line is the 1500 m long transect, a black line represents the 100 m buffer around the transect, the square is 1 km $\times$ 1 km large. B: landscape structure within the 100 m buffer. Yellow – arable fields, light green – grasslands, purple – edge vegetation, dark green – forest, grey – impervious surface (roads and built-in area).



Figure S2. Map of an exemplary grassland site at Wolfbühl, Germany, 750 m a.s.l. A: the red line is the 1500 m long transect, a black line represents the 100 m buffer around the transect, the square is 1 km × 1 km large. B: landscape structure within the 100 m buffer. Light green – grasslands, blue – intensive grasslands, purple – edge vegetation, dark green – forest, grey – impervious surface (roads and built-in area).

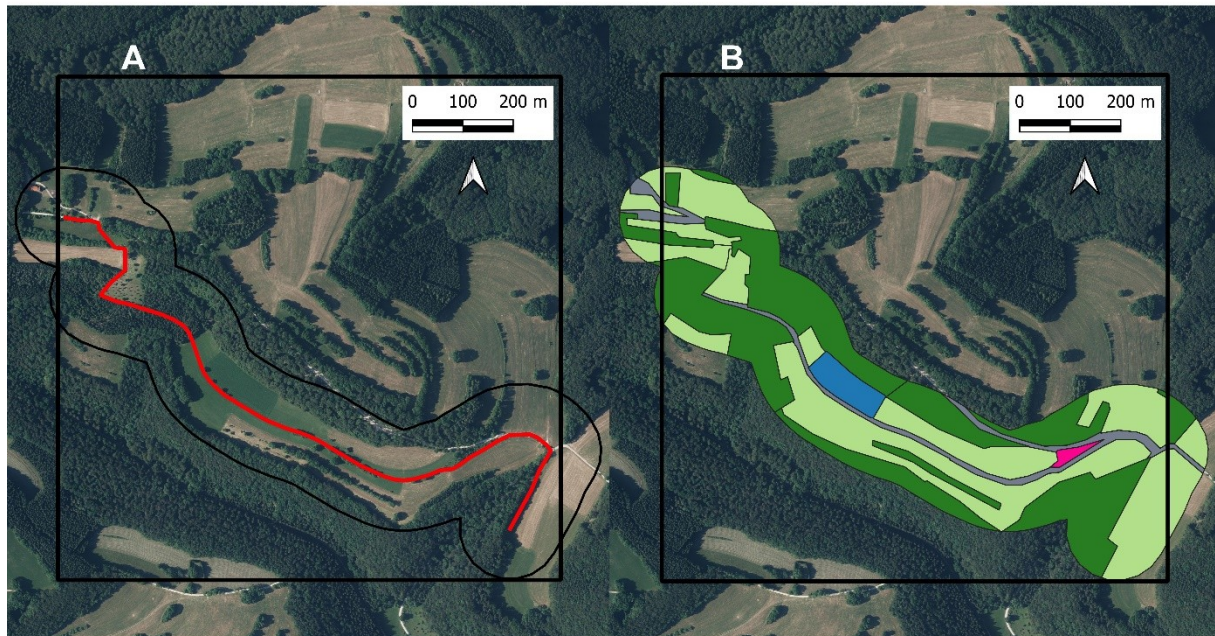


Figure S3. Map of an exemplary nature reserve „Steinriegellandschaft” at Weikersheim, Germany, 305 m a.s.l. A: the red line is the 1500 m long transect, a black line represents the 100 m buffer around the transect, the purple line denotes the nature reserve. B: landscape structure within the 100 m buffer. Yellow – arable fields, light green – grasslands, purple – edge vegetation, dark green – forest, grey – impervious surface (roads and built-in area), white – other.

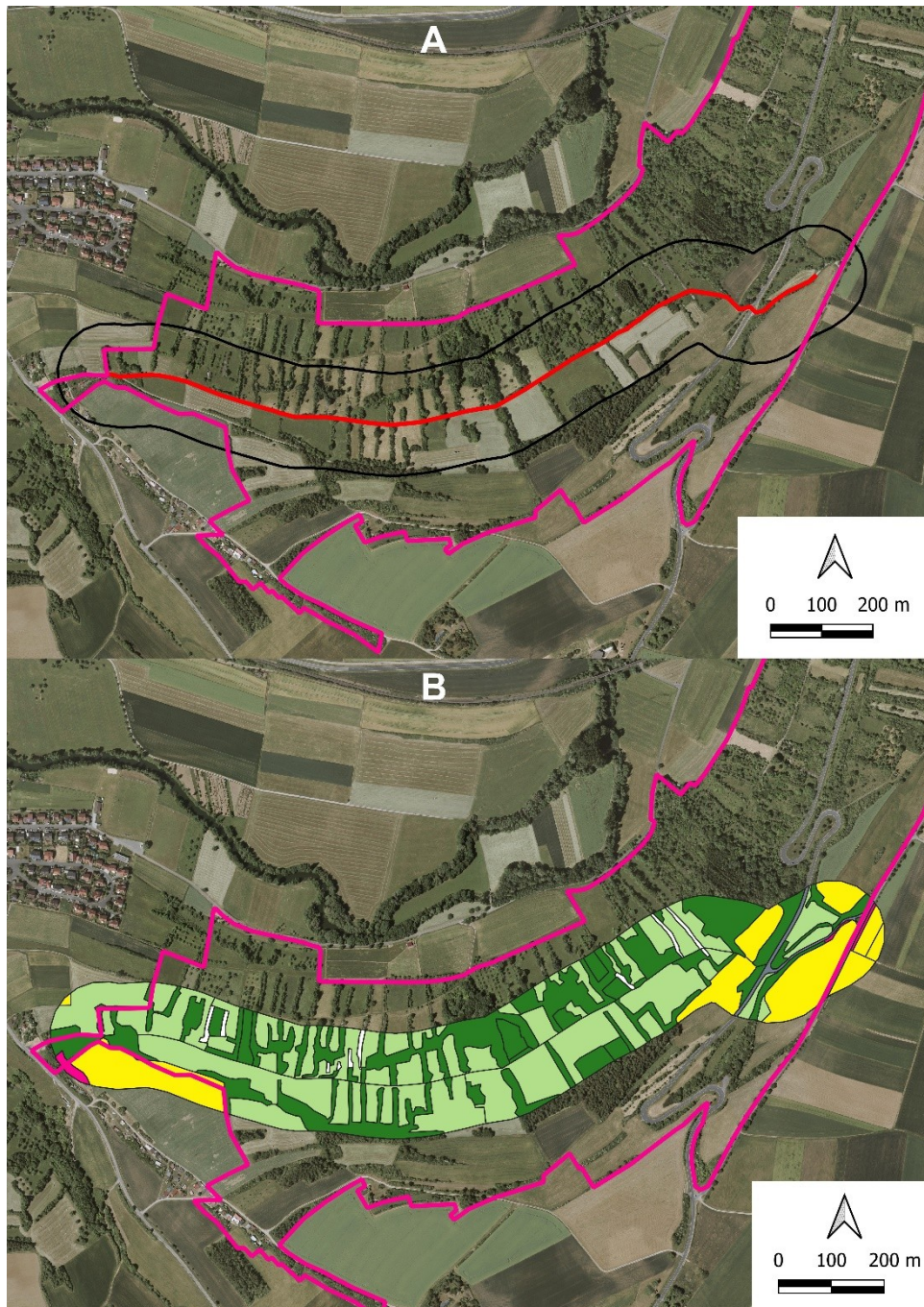


Figure S4. Environmental variables in the three site types. Letters indicate significant differences after multiple comparisons (ANOVA and Tukey post-hoc tests). * Proportion of semi-natural biotopes was square-root transformed. NR: nature reserves.

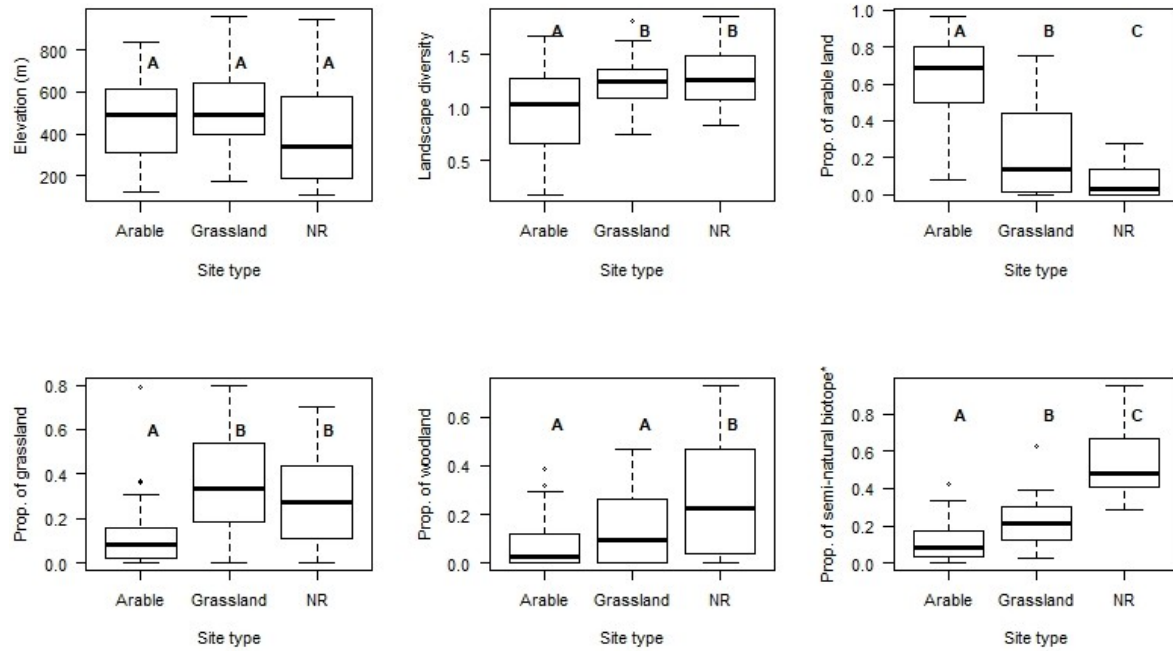
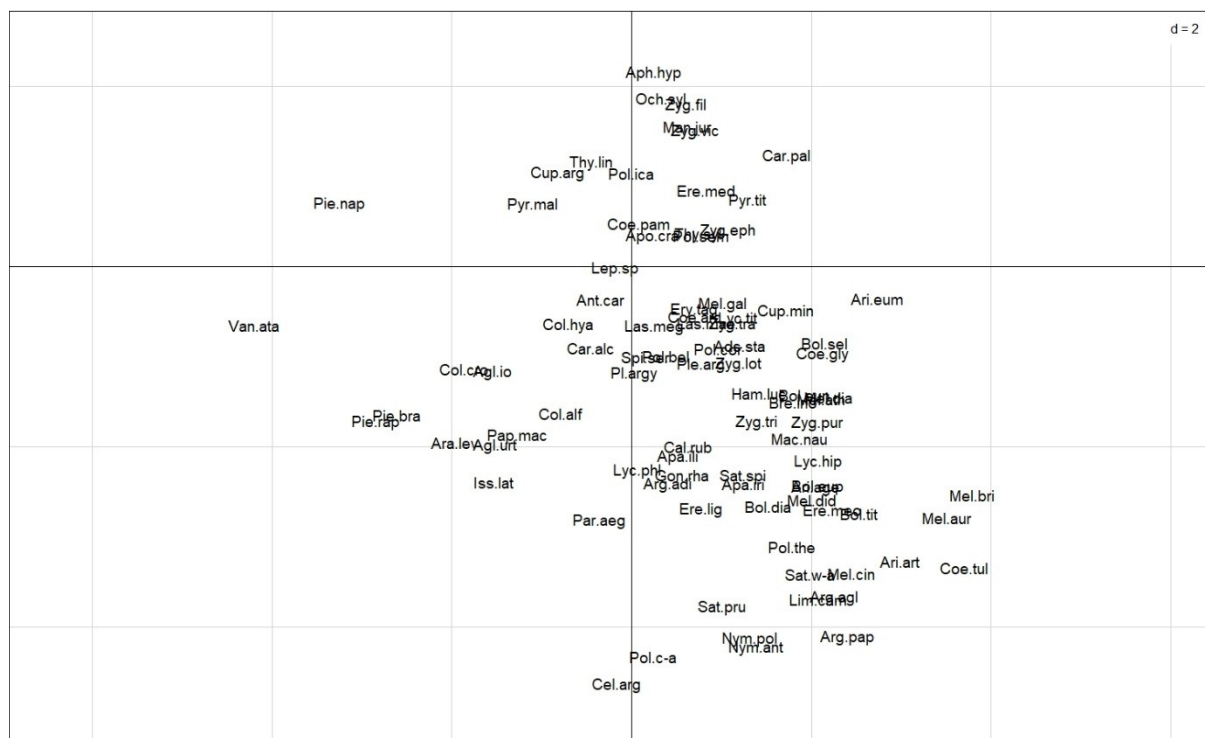


Figure S5. Species scores along the two RLQ axes. Label positions are slightly jittered for better visibility. For abbreviations of species names see SM Table S1.



References

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